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August 12, 2019

Division of Dockets Management (HFA-305)
Food and Drug Administration
5630 Fishers Lane, Room 1061
Rockville, MD 20852

Re: Docket No. FDA-2019-D-0298: Quality Considerations for Continuous Manufacturing; Draft Guidance for Industry.

Dear Sir or Madam:

The Pharmaceutical Research and Manufacturers of America ("PhRMA") is pleased to submit these comments on the Food and Drug Administration's (FDA's or the Agency's) draft guidance entitled "Quality Considerations for Continuous Manufacturing" (Draft Guidance).[1]

PhRMA represents the country's leading innovative biopharmaceutical research companies, which are devoted to discovering and developing medicines that enable patients to live longer, healthier, and more productive lives. Since 2000, PhRMA member companies have invested more than $900 billion in the search for new treatments and cures, including an estimated $79.6 billion in 2018 alone.

PhRMA thanks the Agency for the release of this draft guidance on continuous manufacturing (CM) and appreciates the Agency's efforts to establish a clear and consistent regulatory pathway for this emerging manufacturing technology.[2] Agency guidance and institutionalization of a risk-based approach to develop and evaluate advanced manufacturing technologies will help facilitate adoption and implementation of new technologies that have the potential to improve processes and reduce supply disruptions. PhRMA supports FDA's efforts to facilitate the adoption of CM and other emerging technologies.

In general, PhRMA is supportive of the Draft Guidance; however, we believe the document could benefit from greater clarity on a number of key concepts and that the language can be enhanced to better facilitate the adoption of continuous manufacturing in the biopharmaceutical industry. For the Agency's consideration, PhRMA's comments are detailed below, and a table of proposed line edits can be found in the addendum to these comments.

[1] 84 Fed. Reg. 6403 (February 27, 2019)

[2] PhRMA applauds FDA's efforts to support the modernization of the pharmaceutical manufacturing base through the establishment of the Emerging Technology Program (ETP). PhRMA believes the ETP has significant potential to promote and facilitate the appropriate adoption of emerging technologies and platforms throughout the pharmaceutical industry.

I. General Comments

In general, PhRMA believes the Draft Guidance provides, for drug products, a comprehensive overview of continuous manufacturing and covers the majority of the key topics or areas manufacturers should consider when implementing CM for solid oral dosage forms. However, PhRMA believes some of the language in the Draft Guidance is too rigid and that the Agency's expectations for conducting certain tests during development and process qualification are unnecessarily high. We encourage the Agency to re-assess its expectations to ensure consistency with a risk-based framework for evaluating CM applications.

While the document discusses chemistry, manufacturing, and controls (CMC) topics related to the NDA (e.g., formulation and process development, control strategies) it also contains a number of recommendations with respect to many GMP topics (e.g., validation, cleaning, investigations). CMC and GMP information are interspersed with one another throughout the document, implying that the Agency expects a significant amount of GMP information to be included in the regulatory filing. PhRMA is concerned about the amount of GMP information the Agency appears to be seeking in the NDA and strongly encourages FDA to update the Draft Guidance to clearly distinguish between the information expected in the regulatory filing and what information should be governed by a manufacturer's pharmaceutical quality system (PQS) which can be reviewed during an inspection. PhRMA provides specific comments on elements we believe are appropriate for the PQS, below.

As a general matter, PhRMA believes the Draft Guidance should not include topics that are already addressed in other guidance documents, unless changes to such guidance documents are necessary for their application to CM processes. Additionally, we caution the Agency not to make unnecessary differentiations between batch and continuous manufacturing processes. For example, the objectives of the GMP topics covered in the Draft Guidance are the same for both traditional batch manufacturing and continuous manufacturing. Given that a number of topics and concepts in the Draft Guidance are not unique to CM, PhRMA strongly encourages the Agency to revise the document to include only those elements that are unique to continuous manufacturing processes.

The Draft Guidance also should not implicate the quality of products manufactured by batch processes. For example, stating that CM improves "the overall quality of products"[3] has undertones that suggest there are quality issues for products made by batch processes, which is not the case for the vast majority of batch process-produced products.

PhRMA also strongly encourages the Agency to ensure appropriate flexibility in the recommendations provided by the Draft Guidance. CM is still relatively new to the biopharmaceutical industry; as the industry's experience in CM increases, different strategies and improvements will be identified. To support further adoption of CM technologies, the guidance document should be written to emphasize flexibility, alternate approaches, and ready implementation of post-approval changes.

[3] *See* Draft Guidance lines 65-66

Scope

PhRMA notes that the stated focus of the Draft Guidance is "quality considerations for continuous manufacturing of small molecule, solid oral drug products."[4] PhRMA encourages the Agency to update the title of the document to specify that the Draft Guidance applies only to drug products (and not to drug substances (DS) or other finished dosage forms).

PhRMA also requests that the Agency include examples of solid oral dosage forms other than tablets (e.g., capsules, thin film strips, powders, granules) throughout the document.[5] For example, the section on "Tablet assay and content uniformity by NIR"[6] could substantially benefit from the inclusion of considerations for dosage forms other than tablets. PhRMA believes the generation of an additional Q&A-style document would be helpful in providing specific examples. A Q&A document would allow for high-level recommendations in the Draft Guidance, with more detailed discussion of recommended approaches to implementing CM in specific contexts.

Hybrid Systems

Given that a fully continuous system will not be feasible or practical for all manufacturers, PhRMA encourages FDA to revise the definition of CM in the introduction and in Section V to include a description of hybrid systems and integrated periodic systems. While the Draft Guidance's definition of continuous manufacturing makes passing reference to the possibility of "hybrid" approaches to CM, the guidance's recommendations are primarily directed toward end-to-end processes. Thus, PhRMA recommends that the description clarify whether FDA considers periodic batch operations (e.g., segmented fluid bed dryers, alternating film coaters) to be traditional batch manufacturing operations or continuous manufacturing. Incorporating a clear definition for hybrid systems will reduce industry confusion and better facilitate adoption of CM as well as achieve the benefits that FDA attributes to CM.[7] For example, the guidance should clarify whether the presence of a downstream batch homogenization unit operation with appropriate in-process monitoring would obviate the need for some of the upstream process monitoring and controls.

Expectations for Utilization of Advanced Elements

As mentioned above, and as detailed in the table below, PhRMA believes some of the language in the Draft Guidance is too rigid and that the Agency's expectations for conducting certain tests during development and process qualification are unnecessarily high. The Draft Guidance significantly raises the bar for CM over the current state and could discourage manufacturers from adopting the technology. PhRMA is concerned that the document appears to require the utilization of advanced elements (e.g., QbD, PAT, process monitoring approach,

[4] *See* Draft Guidance lines 15-17

[5] FDA should also consider expanding the scope of the guidance to non-solid oral dosage forms.

[6] *See* Draft Guidance lines 328-341

[7] *See*, e.g., the benefits described in the Draft Guidance at lines 65-77

automation, multivariate trending) in all CM operations. While these approaches may be appropriate and necessary for complex end-to-end systems, implementing them with simple integrated unit operations could have unintended consequences, and would present unnecessary burdens. As written, the Draft Guidance does not clearly contemplate that simple continuous processes could rely on the same development and control strategy approaches as traditional batch manufacturing. PhRMA strongly recommends that FDA soften language throughout the document that implies mandatory incorporation of advanced development and control approaches. Alternatively, the Agency should clarify the conditions under which it considers advanced development and control approaches to be required (if ever) versus those situations in which such approaches are recommended but not required.

International Harmonization

Strategic application and regulation of emerging technologies, including CM, on a global scale will require harmonization. Because adoption of an emerging technology will have a global impact and given FDA's ability to interact and coordinate with other regulatory agencies, PhRMA encourages FDA to continue to proactively support modernization of the pharmaceutical manufacturing base through ICH Q13. PhRMA urges the FDA to share its learnings on CM with other regulators. Shared knowledge will be a key factor in achieving globally aligned text in the ICH Q13 Step 2 document. At the same time, the scope of ICH Q13 is much broader than this Draft Guidance, and PhRMA encourages all ICH stakeholders, including FDA, to ensure that appropriate consideration is given to DS and biologics when drafting the Q13 guideline. While FDA's experience with solid oral dosage forms will be critical in the ICH Q13 process, stakeholders should not adopt a one-size-fits-all approach to CM for all formulations and product types.

II. Defining Batches and Scale for Continuous Manufacturing Processes

Defining Scale

PhRMA believes the Draft Guidance could benefit by introducing the concept of scale earlier in the document. While scale is currently covered in the process validation section,[8] scale may be an earlier consideration. In addition, PhRMA recommends that FDA add a definition of "scale-out" in the Draft Guidance to avoid confusion between descriptions of "scale-up" and "scale-out." We further encourage FDA to ensure that its discussion of scale-up enables batch flexibility for manufacturers. To that end, we propose that scale-up for continuous manufacturing is only related to equipment size and mass flow rate. Run time should be considered a process parameter as opposed to a mode of scale-up.

Defining Batches

PhRMA believes that most simple changes in batch size based on run time should be handled by the PQS, using a science- and risk-based approach, and therefore does not belong in

[8] *See* Draft Guidance lines 489, 498

the regulatory filing. PhRMA encourages FDA to update the Draft Guidance to reflect that batch size/run time does not need to be included in the NDA. PhRMA recognizes that the Draft Guidance includes a provision allowing batch size changes to be managed under the PQS when the manufacturer has prior CM experience. A broader, risk-based approach is more appropriate, including considerations for product and process risks.

Additionally, PhRMA encourages FDA to include recommendations regarding the function of a lot or sub-batch in the context of a CM batch. It would be helpful to include a specific example where it is necessary or advantageous to define a lot or sub-batch for a CM process, and to also include details regarding FDA's expectations tracking lots or sub-batches in the PQS (i.e., lot number management, segregation of material, release of lots vs batches, how to combine lots or sub-batches into a batch).

III. Regulatory Filing Expectations

Capturing Information on Simultaneous Manufacturing Processes in a Single Regulatory Application

PhRMA believes that Section III. G could be enhanced by including a discussion on how both batch and continuous operations can be included in the same application. For a variety of reasons, a manufacturer may choose to produce the same drug product by both batch and continuous processes, simultaneously. PhRMA encourages the Agency to update the Draft Guidance to clearly state that multiple manufacturing processes (i.e., traditional batch manufacturing and continuous manufacturing) can be submitted in the same application and recommends that FDA also include a discussion on allowable variations in formulation and/or unit operations.[9]

Real Time Release Testing

The Draft Guidance states "Although RTRT is not a regulatory requirement for implementation of continuous manufacturing processes, it is encouraged and could be applied to some or all of the finished product quality attributes tested for release of the batch."[10] PhRMA acknowledges and supports that FDA wants to encourage the use of RTRT; however, PhRMA strongly encourages FDA to apply a risk-based approach to the Agency's expectations for RTRT, broadly, and specifically in the context of CM. PhRMA is concerned that language in the Draft Guidance adds unnecessary complexity and burden with respect to RTRT. For example, PhRMA recommends striking the following sentence as it is inconsistent with a risk-based approach to RTRT: "RTRT calculations should also consider the observed variance in CQAs

[9] Elements that could vary for the two control strategies may include slight differences in composition and/or unit operations. The Draft Guidance should explicitly state that variations of aspects such as the formulation, and mode of manufacturing can be utilized as long as the CQAs and QTPP are retained (e.g., same appearance, performance, bioavailability, package label).

[10] *See* Draft Guidance lines 293-295

over a multi-batch campaign to account for both intra- and inter-batch variability."[11] PhRMA does not agree that RTRT calculations (models) should observe such variance in CQAs because RTRT calculations would typically be based on a PAT measurement predicting a certain critical quality attribute (CQA), therefore, batch-to-batch differences in CQAs are not relevant because the prediction is done within each batch. Further, if the model is validated over the appropriate range of the CQA, the batch-to-batch variability should not pose any issue. From another perspective, the relationship between the PAT and CQA may differ for each batch, if it depends on certain additional factors (e.g., a property of a batch-specific input material). However, this should be captured during model development. Any major sources of batch-to-batch differences should be part of the model and minor sources may be ignored so long as precision is sufficient. Once the model is validated, there should not be relevant batch-to-batch differences outside of the parameters considered in the model.

Information for Inclusion in a Regulatory Application vs. Information Managed by the Pharmaceutical Quality System

As noted above, PhRMA is concerned about the extent to which the Draft Guidance appears to suggest that additional GMP data must be included in the NDA. PhRMA strongly encourages FDA to update the Draft Guidance to clearly distinguish between the information expected in the regulatory filing and information that is required as a matter of GMP, but which is governed by a manufacturer's pharmaceutical quality system (PQS) and can be reviewed during an inspection. Specific examples are provided here.

<u>Model Maintenance</u> – Model updates and maintenance should reside within the PQS and be available during inspection but should not require notification, provided the design space does not change.[12]

<u>Sampling Plan</u>[13] – Sampling plans should reside within the PQS and be available for review during an inspection but should not be expected in a regulatory filing. PhRMA strongly recommends removing this language from the Draft Guidance.

<u>Process Monitoring Data</u>[14] – The use of monitoring data for batch release decisions increases the responsibilities for model maintenance. This is suitable if the model maintenance resides completely in the PQS, but if notifications for model updates to FDA are required, it could be a deterrent for companies to use the data for RTRT and batch release. PhRMA strongly recommends removing this language from the Draft Guidance.

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[11] *See* Draft Guidance lines 303-305

[12] "The model maintenance is an example of an activity that can be managed within a company's own internal quality system provided the design space is unchanged." *See* Guidance for Industry, Q8(R2) Pharmaceutical Development, at p. 14 (section 2.6).

[13] *See* Draft Guidance lines 212-217

[14] *See* Draft Guidance lines 225-233

Equipment & Cleaning[15] – Considerations with respect to manufacturing equipment are strictly GMP. This is an unusual inclusion in the Draft Guidance as batch processes (especially those with PAT) have the same or similar issues as CM equipment. PhRMA strongly recommends removing this section from the Draft Guidance.

System Integration, Data Processing, and Management[16] – PhRMA recognizes that for API, the processes may be manual, and that automation may be required, however it is not essential as suggested by the wording of this section. We encourage the Agency to describe in the guidance considerations with respect to system integration, data processing, and management that are unique to CM, but PhRMA maintains that this information should be maintained by the PQS. The guidance should explicitly state that this information is not expected in a regulatory filing.

Training[17] – Training is a basic GMP requirement for all methods of manufacturing and should not be discussed in the Draft Guidance. Given that the document does not provide any justification or explanation for why training requirements would be different for batch and continuous processes, PhRMA strongly recommends removing this section and any references to training from the Draft Guidance.

Post-approval Filing Strategies for Scale-Up

PhRMA strongly supports the concept of offering regulatory flexibility "to manage […] scale-up […] by the facility's PQS without a supplement or comparability protocol."[18] PhRMA recommends that FDA update the language in this section to specifically include consideration of manufacturer prior knowledge of product and process risks in assessing which firms may utilize/benefit from this regulatory flexibility, rather than solely relying on a firm's previous experience with a "suitably similar continuous manufacturing"[19] process.

Regulatory Flexibility for Post-Approval Changes

The Draft Guidance states that the Agency expects "operational flexibility may decrease the need for some post-approval regulatory submissions."[20] PhRMA suggests that FDA update the document to include specific language and examples detailing how particular control strategy approaches can reduce post-approval regulatory submissions.

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[15] *See* Draft Guidance lines 368-417

[16] *See* Draft Guidance lines 419-475

[17] *See* Draft Guidance lines 614-619

[18] *See* Draft Guidance lines 692-693

[19] *See* Draft Guidance line 691

[20] *See* Draft Guidance lines 74-75

IV. Process Validation and Process Verification

Continuous Process Verification

PhRMA believes the Draft Guidance could be greatly enhanced by the inclusion of discussion and examples of Continuous Process Verification in the context of CM as an alternative for process validation, as discussed in ICH Q8 and other supportive ICH documents.

PhRMA encourages the Agency to allow for flexibility with respect to process validation. In addition to the concepts included in the section on PPQ, PhRMA encourages FDA to consider incorporating the Continuous Process Verification definition in ICH Q8: "An alternative approach to process validation in which manufacturing process performance is continuously monitored and evaluated." Where advanced technologies enable continuous monitoring and real time evaluation, an alternative approach to traditional process validation (PPQ) may be allowable. Continuous Process Verification can be used in addition to, or instead of, traditional process validation. PhRMA believes this strategy fits within the definition of a lifecycle approach to validation and urges FDA to consider Continuous Process Verification as an acceptable alternative to PPQ, provided that certain conditions are met.[21]

PPQ Considerations

PhRMA thanks the Agency for including a section in the Draft Guidance on process qualification and offers the following comments.

The Draft Guidance states "The PPQ protocol should be designed to assess robustness with respect to the known sources of variability including those unique to continuous manufacturing processes."[22] The language in the Draft Guidance appears to imply that manufacturers would need to introduce by purpose some variability within the process (e.g., induce feeder disturbances). PhRMA recommends that FDA revise this language to align with the usual practice of running PPQ batches as target batches.[23]

In addition, the document indicates that "manufacturers should establish measures of process stability and associated acceptance criteria as part of the PPQ protocol."[24] More guidance may be needed on the concrete framework required by FDA. Batch-to-batch reproducibility may not be feasible in state of PPQ due to low number of batches typically used at this stage. Inter-batch variability (for batch processes) has been a matter of CPV3a, not PPQ.

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[21] In addition to Continuous Process Verification, PhRMA encourages the Agency to also consider Continuous Monitoring as an alternative to traditional process validation.

[22] *See* Draft Guidance lines 525-527

[23] Consistent with this feedback, PhRMA also recommends that FDA strike or revise lines 553-554 of the Draft Guidance which state "The magnitude and duration of variability for process parameters and quality attributes should be evaluated as part of the PPQ protocol, and should be justified."

[24] *See* Draft Guidance lines 531-532

PhRMA is concerned that this language signals a shift in FDA's thinking for both batch and continuous processes, and requests clarity from the Agency on this issue.

PhRMA is concerned by the statements in lines 538-549 of the Draft Guidance. This paragraph, as written, is not consistent with the general philosophy of CM of flexible batch size and instead supports a single, validated batch size. PhRMA does not believe that campaign length should be a part of the PPQ discussion and encourages the FDA to update this section to clarify that an initial PPQ campaign that spans a range of process parameters, including run time, would enable the manufacturer to operate within that demonstrated range. The updated language should also state that the potential risks associated with an increase in run time may be assessed in a risk assessment as a part of the application. Additionally, FDA should remove the language requiring PPQ to capture variability associated with campaigning, as this is not an expectation for batch processes.

V. FDA-Requested Feedback

In response to FDA's request in the Federal Register Notice for feedback on "additional issues for consideration," PhRMA offers the following comments.

Data storage and handling from process analytical technology systems

PhRMA would like the Agency to consider in what cases storage of reduced data, or only storage of analyzed data, would be appropriate.

Potential approaches for situations where direct attribute measurement is not possible (e.g., low-dose compounds)

PhRMA recommends that Agency research consider surrogates for direct measurement of active ingredients, such as process modeling and simulation.

Risk-based reporting of routine model maintenance and updates

As noted above, PhRMA believes that model updates and maintenance should reside within the PQS and be available for review during an inspection but should not require notification.

VI. Conclusion

PhRMA thanks the Agency for the opportunity to comment and welcomes further discussion with FDA regarding quality considerations for continuous manufacturing and PhRMA's specific comments on this Draft Guidance.

Respectfully submitted,

/s/

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ADDENDUM

Line Number(s)	Current Language	Proposed Language	Justification
Title	"Quality Considerations for Continuous Manufacturing Guidance for Industry"	Quality Considerations for Solid Oral Drug Product Continuous Manufacturing	The introduction on lines 15 – 21 clearly indicates this guidance is specific to small molecule, solid oral drug products. To that end, it would be helpful to have this scope identified in the title.
23-28	"However, this guidance is not intended to provide recommendations specific to continuous manufacturing technologies used for biological products under a BLA."	Either remove final sentence entirely or modify, as shown below. *Proposed modification*: However, this guidance is intended to provide recommendations specific to continuous manufacturing technologies used for small molecule, solid oral drug products only.	This statement is unclear with respect to the scope of the guidance. The final line in the first paragraph specifically excludes "biologic products under a BLA", while not excluding all other examples in lines 23-26. It could cause confusion if biologic products are specifically listed as excluded and other drug substance and drug products discussed in lines 23-26 are not also listed.
30-35	"For purposes of this guidance, FDA considers "continuous manufacturing" to be a process in which the input material(s) are continuously fed into and transformed within the process, and the processed output materials are continuously removed from the system. Although this description can be applied to individual unit operations or a manufacturing process consisting of a series of unit operations, as described in this guidance, continuous manufacturing is an integrated process that consists of a series of two or more unit operations."	Add to definition of continuous manufacturing a description of hybrid systems including batch elements. Include if periodic batch operations (e.g., segmented fluid bed dryers, alternating film coaters) are considered continuous or batch.	The definition of continuous manufacturing in the introduction and in Section V should include a definition of hybrid systems and integrated periodic systems. A clear definition will improve industry understanding of FDA's expectations, increase adoption of CM, and facilitate consistent application of the recommended approaches.
39-40	"These considerations include process dynamics, batch definition, control strategy, pharmaceutical quality system,	These considerations include process dynamics, batch definition, control strategy, pharmaceutical quality system, scale-up, stability, and bridging	The material from the process is bridged, not the process itself.

Line Number(s)	Current Language	Proposed Language	Justification
	scale-up, stability, and bridging existing batch manufacturing to continuous manufacturing."	of material made from existing batch manufacturing to that made from continuous manufacturing.	
73	"[…] scale-up, scale-down, and scale-out […]"	Consistent with our above comments (page 4), PhRMA recommends that FDA define "scale-up," "scale-down," and "scale-out" in the glossary of the guidance.	Defining terms will ensure stakeholders understand the distinction between scale-up, scale-down, and scale-out as used in the guidance.
91-92	"Continuous manufacturing processes are dynamic systems unlike batch manufacturing processes."	Delete text.	This statement is incorrect and should be removed. Batch manufacturing is dynamic as compositional profiles and often process parameters change with time.
112-114	"The shape of the RTD reflects the degree of axial dispersion or back mixing within that system, which affect the propagation of disturbances, material traceability, and the control strategy (e.g., material diversion and sampling frequency)."	Replace with "The shape of the RTD reflects the degree of axial dispersion or back mixing within that system, which affects the propagation of disturbances and material traceability. The control strategy should consider the RTD characteristics."	The RTD does not directly affect the control strategy.
163-165	"To maintain a process within a state of control during continuous operation, detect temporary process disturbances, and segregate the resulting nonconforming materials from the system, manufacturers should increase the use of in-process control strategy elements."	Replace with "Manufacturers may increase the use of in-process control strategy elements to maintain a process within a state of control during continuous operation, detect temporary process disturbances, and segregate (confirmed) nonconforming materials from the system."	PhRMA agrees that nonconforming material should be diverted. However, depending on the process design and controls, nonconforming material may not necessarily be produced during start-up, shutdown, or process disturbances. Temporary process disturbances may trigger material diversion depending on various factors such as magnitude or system dynamics, but subsequent evaluations may demonstrate that this material is still acceptable for use. We request that statements be clarified that nonconforming material should be diverted, but without unilaterally linking process disturbances to the expectation that material must be rejected.

Line Number(s)	Current Language	Proposed Language	Justification
170	"*1. Input Material Control*"	*1. Input Materials*	Sensitivity analysis will be key to understanding controls required on a specific material attribute. If the system is not sensitive to an attribute, then no additional control may be necessary.
180-185	"Suitable risk analyses, experimental investigation, and/or modeling and simulation should be considered throughout the life cycle of the product, including during pharmaceutical development, to evaluate potential impact of material attributes (e.g. particle size distribution and density of the active pharmaceutical ingredient (API) and excipients) on the material flow properties, process dynamics, and quality of a final product over the period of an intended production run."	Suitable risk analyses, experimental investigation, and/or modeling and simulation should be considered throughout the life cycle of the product, as needed, including during pharmaceutical development, to evaluate potential impact of material attributes (e.g., particle size distribution and density of the active pharmaceutical ingredient (API) and excipients) on the material flow properties, process dynamics, and quality of a final product, and where appropriate, over the period of an intended production run.	It is not clear why it is necessary to conduct risk analyses, experimental investigation and/or modelling simulation over the entire lifecycle of the product. Suggest that this is done only "as needed." It is not clear what is meant by "over the period of an intended production run." This may imply these studies need to be conducted for all projects. This may only need to be evaluated for some projects, depending on the risk assessments and the specifics of the continuous manufacturing physical build, the manufacturing platform, and the specifics of the drug substance/formulation in question.
192-194	"Manufacturers planning to switch from batch to continuous manufacturing should take a similar approach to re-evaluate the existing specification(s) for raw materials and their use in a particular continuous process design."	Add "Specifically, manufacturers should evaluate how any material attributes not included in the existing excipient or material specifications for batch manufacture could affect the continuous process or product quality. Manufacturers should consider implementing additional or different controls for additional material attributes identified as part of this evaluation."	Provides additional clarity and guidance to manufacturers.
200	"[…] utilization of PAT tools generate real time information on process parameters […]"	...utilization of PAT tools may be used to generate real time information…	Provides clarification that the use of PAT is recommended but not required.

Line Number(s)	Current Language	Proposed Language	Justification
206	"Variables being monitored at appropriate locations in the process, such as:"	Variables that may be monitored at appropriate locations in the process, may include:	Provides clarification that these are recommended, not required, variables.
209	"Input and in-process material attributes, and"	In-process material attributes, and	Monitoring material attributes may not be necessary, especially if material characterization studies have identified CMAs, and controls are in place (e.g., on the purchase specification), that ensure the critical attributes are within an acceptable range that assures proper performance.
212-217	"Sampling plan, including: o Sampling locations, o Sampling or measurement frequency, o The sample size to be taken and measured, and o Statistical criteria appropriate for use to evaluate the process monitoring data."	Delete text. Alternatively, revise line 204 to read "A process monitoring approach may include the following elements" such that the sentence is less prescriptive.	Sampling plans belong in the PQS, not the NDA.
222	"Multivariate or process model, and"	Multivariate or process model, or	This text is very prescriptive as written. "And" makes all things inclusive; "or" implies any or all.
225-233	"Intended use(s) of process monitoring data, such as: o Supporting other control strategy elements (e.g., active process control, material diversion, RTRT, batch release), o Evaluating process and equipment performance as part of process development, during manufacturing, and to facilitate continued process verification,	Delete text. Alternatively, revise line 204 to read "A process monitoring approach may include the following elements" such that the sentence is less prescriptive.	All things presented in these lines belong in the PQS and should not be submitted in the NDA.

Line Number(s)	Current Language	Proposed Language	Justification
	o Ongoing monitoring of a process to confirm that it remains under a state of control, and o Additional elements of the Pharmaceutical Quality System."		
243-247	"The process monitoring approach selected should include alternative or additional quality controls to mitigates the risks to product quality posed by these scenarios."	Replace with, "The process monitoring approach should mitigate the risks to product quality using additional or alternative quality controls as determined by the sponsor."	The approach should be risk-based; if the risk is unlikely to occur, is easily detected, and would not affect SISQP, then additional or alternative controls may not be appropriate. Sponsor is responsible for making this determination.
254-255	"The establishment of appropriate limits (e.g., alarm or action limits) is also important for robust control."	Replace with "Usable ranges of process parameters and quality attributes should be established, and alarm and action limits may be considered practical to assure a stable process."	Proposed language in the guideline suggests that action limits or alarm limits may be required. This may not be required in all cases. (May prevent excess manipulation of the process due to measurement uncertainty.)
255-257	"The limits of acceptability for controls that ensure monitored critical process parameters and critical material attributes stay within desired ranges should be specified in the regulatory submission."	Replace with "The limits of acceptability for controls, or other control criteria that ensure monitored critical process parameters and critical material attributes stay within desired ranges, should be specified in the regulatory submission."	This section is too prescriptive with respect to limits of acceptability for controls needing to be specified in the regulatory submission.
262	"However, the manufacturing process will include […]"	However, the manufacturing process can include…	It is not a given that the manufacturing process *will* include periods when nonconforming material is produced, therefore "will" should be changed to "can."
293-295	"Although RTRT is not a regulatory requirement for implementation of continuous manufacturing processes, it is encouraged and could be applied to some or all of the finished product quality attributes tested for release of the batch."	Add "Traditional release based on sample testing may be accepted."	In this document one gets the impression that PAT (=on-line measurements) equates to RTRT (NIR=PAT=RTRT). This should be more differentiated. On-line tools can be employed for material diversion or to prove homogeneity of the run, based on which a release by means of end-product testing might be justified.

Line Number(s)	Current Language	Proposed Language	Justification
			RTRT goes one step further using the on-line data for release purposes which for certain CQAs might be very challenging, e.g. dissolution of a MR tablet or degradation products. The document does not, but should, explicitly state and explain what the alternative to an RTRT approach would look like.
299-306	"The selected sample size or frequency should be representative of the batch and the approach should be justified using an appropriate statistical approach."	Replace with "The selected sample size or frequency should be representative of the batch and the approach should be justified."	Recommend clarification of 'appropriate statistical approach'. Sampling frequency is largely based on system dynamics. Process dynamic and appropriate diversion of non-conforming material play a role in sampling scheme. Time series sampling is fundamentally different than a statistical sampling plan since the sampling interval correlates to the deviation between previous and next samples. A statistical sampling plan, as in batch processes, can be used for CM in a classical sampling and release testing scheme. However, this is not fully applicable to RTRT.
299	"[...] inline sampling. The selected sample size or frequency [...]"	Replace with "...in-line testing. The selected sample size or testing frequency..."	For in-line tests, samples are not taken.
301-303	"For data collected at high frequency, statistical methods for large sample sizes should be applied to provide improved characterization of a batch."	For data collected at high frequency, statistical methods for large sample sizes should be used to assure the quality of the batch relative to specific quality attributes (e.g., content uniformity).	It is not clear what is meant by large samples to "characterize the batch." Large sample sizes generally provide a more representative characterization (for example, in calculating means and standard deviations). It seems as if the implication is that large sample procedures for assuring the quality of the batch are appropriate.
305-306	"Furthermore, procedures should be developed to establish a plan for RTRT	Replace with "Alternative procedures should be developed in the event of failure of PAT used for RTRT. Off-line methods are reference methods	Off-line test methods are the basis for qualification of PAT analysis.

Line Number(s)	Current Language	Proposed Language	Justification
	to address potential gaps in PAT data (e.g., failure of the PAT equipment).”	to which PAT is qualified. Off line methods provide the analytical testing to be used for stability testing, market surveillance and verification.”	
311-312	“[…] multivariate models to predict dissolution for release and calibration models associated with NIR procedures that are used for content uniformity and assay release testing.”	Replace with “…multivariate models and other suitable methods to predict dissolution for release and calibration models associated with NIR, FT-iR and/or Raman procedures that are used for content uniformity and assay release testing.”	Other spectrometric methods are applicable beyond NIR.
322-323	“The impact of any unique identifiers such as embossing and sample orientation on the test method should be examined.”	Replace with “The impact of any unique identifiers such as sample orientation on the test method should be examined.”	If the ID test is performed on the product, robustness against changes in embossing should be demonstrated in method validation. Recommend removing embossing requirement. The purpose is not clear.
328	“Tablet assay and content uniformity by NIR”	Replace with “Tablet assay and content uniformity by non-destructive methods”	Suggest including reference to relevant FT-IR and Raman methods; while there is a NIR guidance, there are relevant examples of the other techniques that should be at least included as alternative approaches.
332	“[…] the finished tablet […]”	Replace with “…the finished tablet, capsule, powdered dosage forms (e.g., powders for suspension), or other solid oral dosage forms…” or “…the finished dosage form…”	The guidance applies to solid oral dosage forms. Tablets are only one example of solid oral dosage forms.
334-336	“The NIR measurement of active concentration in the tablet should account for tablet weight […]”	Replace with “The non-destructive method should consider possible mass variations, if needed.”	Suggest deleting the requirement to account for tablet weight since this is not always required and technically not always feasible.
337-338	“The PAT tool used for RTRT should be validated against the offline analytical	Replace with “The PAT tool used for RTRT should be validated against an appropriate reference based on intended use.”	There are situations in which a PAT tool could be validated against something other than an “offline analytical method.”

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	method (e.g., High-Performance Liquid Chromatography)."		
339-341	"The calibration model associated with the NIR method should be adequately developed and validated over the proposed operating ranges for commercial production."	Replace with "The calibration model associated with the spectroscopic method should be adequately developed and validated over the proposed operating ranges for commercial production and consider the permitted ranges of any critical material attributes."	The spectrometric model can be influenced by material attributes which should be considered in the validation of the model
351-352	"[…] as well as demonstrate specificity (e.g., capability of detecting nonconforming product)."	Delete text.	This language appears to imply that manufacturers should either deliberately produce and/or look for some population of nonconforming behavior in order to demonstrate that the model is capable of detecting nonconforming product. Such an expectation from the Agency would be unnecessary to demonstrate the capability of the model.
481	"The guidance for industry Process Validation: General Principles and Practices and ICH Q8, Q9, and Q10, is applicable to continuous manufacturing processes."	Replace with "Validation applies where no real-time control is in place and offline testing is used to manage quality. If qualified real-time data acquisition and evaluation is used to manage the quality of the product and adjustment in real time is possible, continuous process verification can be used in place of traditional process validation."	If real-time data is available to describe the process, process validation may become obsolete where quality is assured in real time. If this is not the case, and the classical test and release principle is applied, then traditional validation approach applies.
481-484	"For these types of manufacturing processes, the ability to evaluate real time data to maintain operations within established criteria to produce drug products with a high degree of assurance of meeting all the attributes they are intended to possess is an integral element of process validation."	Delete text.	The text contradicts statements in lines 293-295 depicting that real-time release testing is optional: "Although RTRT is not a regulatory requirement for implementation of continuous manufacturing processes, it is encouraged and could be applied to some or all of the finished product quality attributes tested for release of the batch."

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			Please delete the language in lines 481-484 and clarify that real-time controls are not a validation (regulatory) requirement.
513-521	"Qualification of the integrated equipment and automated control systems is essential for ensuring the performance of a continuous process. Given the interrelated nature of the integrated equipment, process design, and control strategy, the first component described in stage 2 of the *Process Validation* guidance, Design of Facilities and Qualification of Utilities and Equipment, may often be more appropriate to perform in stage 1. Additionally, information on the equipment and automation system performance and its variability will inform the design of the PPQ protocol. Because the reliable performance of equipment and automation is critical for PPQ, manufacturers should evaluate whether they have sufficient experience with the fully integrated continuous manufacturing process before initiating PPQ."	PhRMA proposes to expand the current content to reflect that any PAT or process models, disturbance detection methods, diversion controls, and start-up/shutdown controls may be qualified in Stage 2a to demonstrate the qualified state of the integrated system. PhRMA also proposes that wording be added to clarify that these components would not need to be demonstrated, or challenged, multiple times during the Stage 2b process validation.	PhRMA agrees that the role of Stage 2a qualification is expanded for continuous processes relative to batch. Equipment and automation are listed as examples, however other aspects of the integrated system such as any PAT or process models, disturbance detection methods, diversion controls, and start-up/shutdown controls may also be qualified in Stage 2a to demonstrate the qualified state of the integrated system. As such under this approach, once qualified these components would not need to be demonstrated, or challenged, multiple times during the Stage 2b process validation.
517-519	"Additionally, information on the equipment and automation system performance and its variability will inform the design of the PPQ protocol."	Replace with "Additionally, information on the equipment and automation system performance and its variability will inform the risks and the control strategy."	Unclear how "information on the equipment and automation system performance and its variability" informs the PPQ protocol; this should be evaluated during development.
579-580, 604-605	579-580 – "Quantitative and statistical methods, including multivariate approaches, whenever appropriate and feasible;"	For lines 579-580, delete "including multivariate approaches, whenever appropriate and feasible"	Multivariate analysis and trending represents a new and higher-level requirement for continuous manufacturing; they are not always

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	604-605 – "PPQ protocol and continued process verification approach, including process robustness, actual yield, and multivariate tracking and trending"	For lines 604-605, delete "and multivariate tracking and trending"	needed to sufficiently analyze process performance and product quality.
600-601, Page 20 – eCTD Location 3.2.P.2.3 (4th bullet, 4th sub-bullet)	600-601 – "In-process material diversion strategy, including the criteria for rejection of the entire batch" Page 20 – "Description of the current criteria for rejection of the entire batch"	For lines 600-601, replace with "In-process material diversion strategy, including the current investigation criteria for determining whether an entire batch should be rejected." For Page 20, replace with "Description of the current investigation criteria for determining whether an entire batch should be rejected."	Rejection of an entire batch should be based on the results of a quality investigation assessing the process disturbance(s), process controls, and data confirming product quality, not necessarily pre-determined criteria for batch rejection. We believe that companies should be able to leverage the rigorous controls and monitoring in place for continuous manufacturing processes to divert impacted material and then leverage PQS to evaluate the final material based on established quality measures (e.g., PARs, IPCs, CQAs associated with the final drug product, etc.) to make the batch disposition decision.
642	"Equipment "dead zones," […]"	Please add language defining what an equipment "dead zone" is or remove this text.	Unclear whether this term is widely understood or commonly defined. Clarification strongly suggested.
707-709	"Alternatively, stability samples could be obtained from a single continuous manufacturing campaign where manufacturing variability is captured (e.g., by introducing different batches of input material(s) in a sequential manner)."	Consider harmonizing with all guidance areas requiring characterizing within- and between-batch variability.	Continuous manufacturing offers the unique opportunity of studying various material, equipment, and operator conditions with the same run and batch. Appropriate material traceability and sampling controls are the critical enablers in this case, which allow new development and qualification modes (experimentation within-run). The stability section correctly describes this opportunity, in contradiction to several earlier statements describing within- and between-batch variability which relates more to

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			traditional batch processing (lines 188, 493, 531). Please consider a harmonized approach across the document.
720-721	“A change from batch to continuous manufacturing is a change in the scientific operating principle, and it likely results in changes […]”	Replace with “A change from batch to continuous manufacturing may be a change in the scientific operating principle, and it may result in changes …”	The impact of a change from batch to continuous will depend on the unit operation and should be evaluated by a risk-assessment. The language in the Draft Guidance should not be so rigid.
729-732	“In the cases where the continuous process may be based on the same unit operations and formulation as used for the batch process, the risk of change to product quality attributes (e.g., polymorphic form, dissolution, impurities, and stability) may be low and demonstration of in vitro equivalence may be sufficient to support such a change.”	Replace with “In the cases where the continuous process results in similar product quality (e.g. polymorphic form, dissolution, impurities, and stability) demonstration of in vitro equivalence may be sufficient to support such a change from batch to continuous, regardless of the potential unit operation and/or formulation differences. Additionally, the required supporting stability data should be commensurate with the level of risk associated with the changes to the formulation and/or process.”	PhRMA agrees that an evaluation of the transition from batch to continuous manufacturing should include a comparison of individual unit operations, process parameters, equipment, CQAs, and elements of the control strategy. We also agree with the supposition maintaining the same unit operations and formulation can enable in vitro data only to support a change to a continuous manufacturing process. However, the transition should not be limited to maintaining the same unit formula and similar operating principles but should be evaluated using the appropriate tools to assess the risk of transition since the risk of change to product quality attributes (e.g., polymorphic form, dissolution, impurities, and stability) may be low and demonstration of in vitro equivalence may be sufficient to support a change in the formulation and manufacturing process. Demonstration of in vitro equivalency may be supported by comparative batch data, including (not limited to) physicochemical properties (e.g., polymorphic form and particle size), impurity profiles, drug release profiles,

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			and bridging stability data. PhRMA is also in agreement and is appreciative of the Agency's willingness to discuss the proposed changes and the bridging strategy to obtain feedback and alignment prior to conducting the studies to transition from a batch process to continuous process even if there are changes in formulation and manufacturing platform to transition away from the batch process.
Table 3.2.P.2.3 Pg. 20	"Description of the current criteria for rejection of the entire batch"	Remove this text or replace with "Description of criteria to accept a batch"	The NDA for a batch process does not currently require descriptions of criteria to accept or reject a batch, so it is not seen as necessary for CM. If anything, then it should be a positive statement of releasing a batch based on results of in process data and end-product testing.
Table 3.2.P.2.3 Pg. 20	-	Section 3.2.P.2.3 should clarify that only a **high-level overview** of the material strategy and material collection and diversion strategy be included in the dossier.	Material traceability strategy and collection and diversion strategy are elements of the PQS and a detailed description should be required to be included in the regulatory application.
Table 3.2.P.3.4 Pg. 21	"Justification of sampling strategies for finished product testing […]"	Remove this requirement or replace with "Justification of sampling strategies for in-process testing (e.g., when used in lieu of finished product testing)"	Sampling for finished product testing would be described in P5 instead.
Table 3.2.P.3.4 Pg. 21	"Development and supporting data for PAT methods and models"	Delete text.	Duplication of information related to PAT methods & models in P.2.3 and P.3.4. Development & supportive data for PAT methods & models should be in P2.3.
Table 3.2.P.3.4 Pg. 21	"Description of advanced process control approaches (e.g., feedback, feedforward, model predictive)"	Delete text.	The same information is provided already in 3.2.P.2.3 (as part of development discussion) and in 3.2.P.3.3 as part of the process flow and control. It should not be repeated. Please clarify if different information should appear in P.3.3 and P.3.4.